



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KONKAN FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

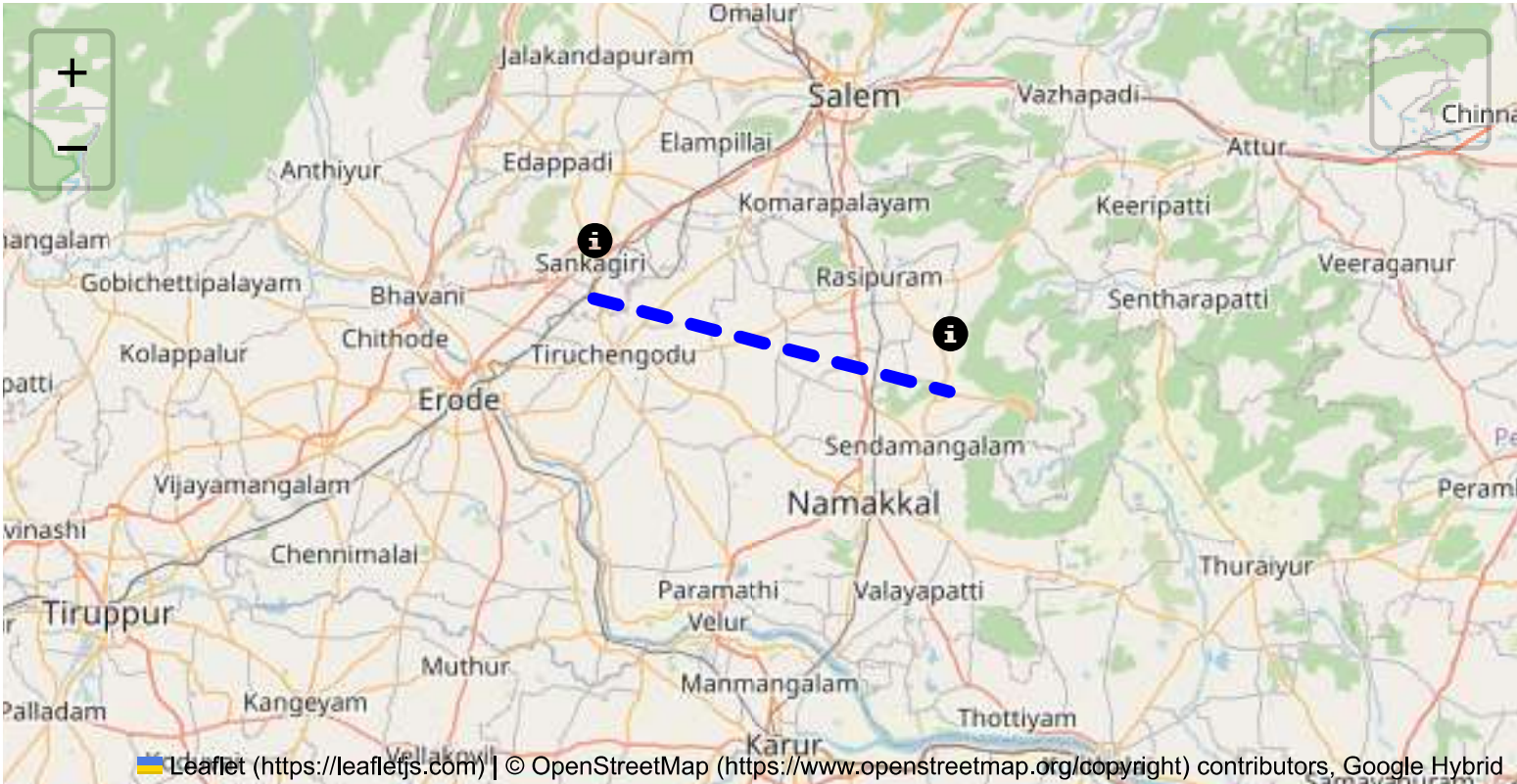
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

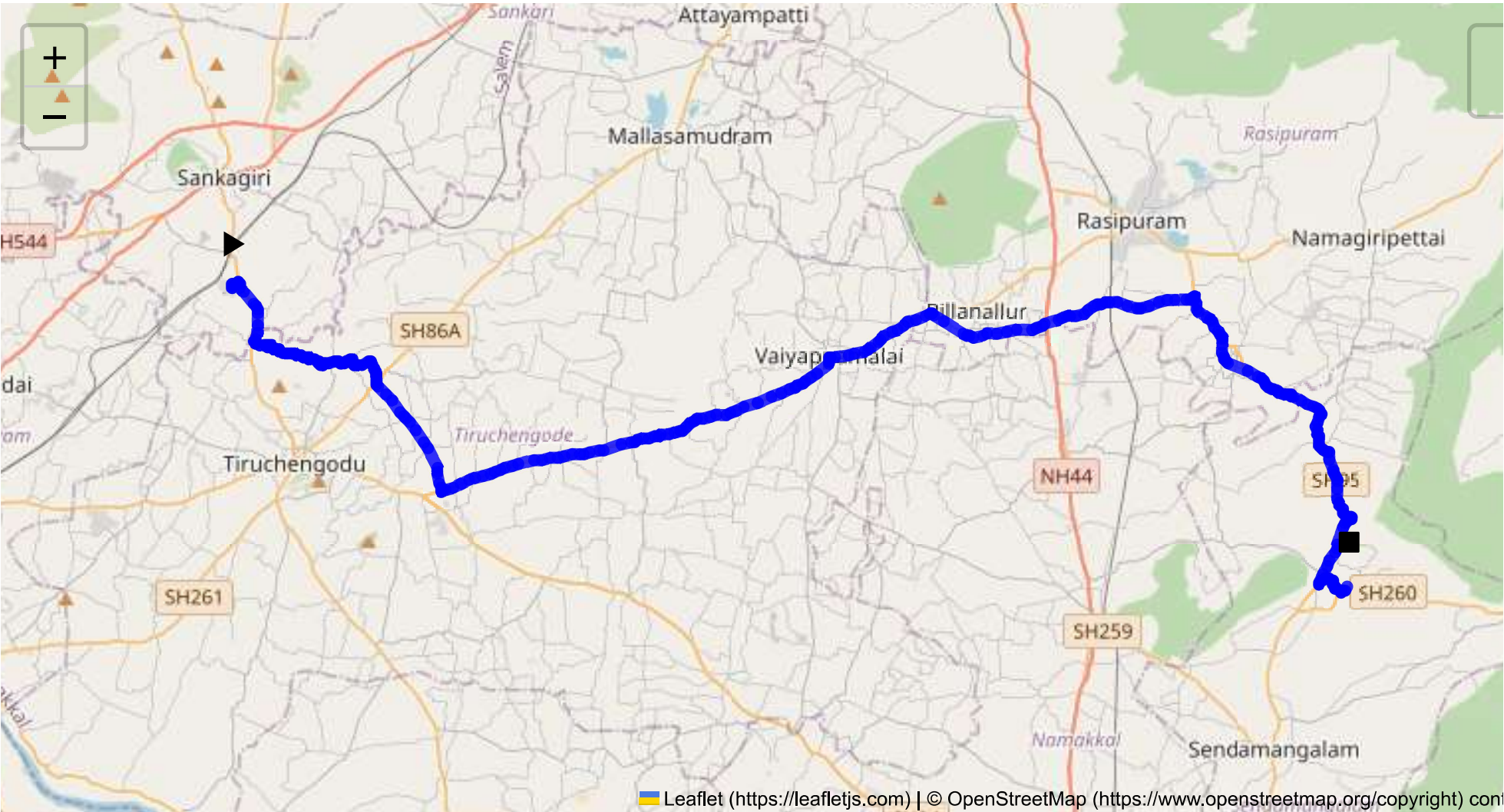
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 60.68 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (11.336364, 78.260701)



Welcome to the Journey Risk Management Study

Route Analysis from Sangagiri to Nachipudhur, Tamil Nadu

1. Overview of the Route Map:

The route primarily follows a network of local and state highways, starting from Sangagiri, proceeding through urban and semi-urban regions, and concluding in Nachipudhur. Key locations along the way might include towns and road junctions that serve as important landmarks.

2. Typical Weather Conditions:

The region generally experiences a tropical climate. The primary weather-related hazards are:

- The monsoon season (June to September) can bring heavy rains, potentially causing flooding and slick roads.
- Summer months might have high temperatures exceeding 40°C, leading to asphalt softening. It is crucial to monitor weather forecasts regularly to mitigate these risks.

3. Traffic Patterns:

- Peak hours typically occur from 8:00-10:00 AM and 5:00-7:00 PM. Congestion is likely around major town intersections and entry/exit points of Sangagiri and Nachipudhur.
- Local markets and schools can cause sporadic congestion.

4. Assessment of Road Quality and Infrastructure:

- State highways are generally maintained but may have sections with potholes or uneven surfaces.
- Some rural roads might be narrower with less lighting, creating visibility issues, especially at night.

5. Suggestions for Alternative Routes for Emergencies:

- In case of major blockages, consider using NH-544, which connects to other highways leading to the destination, although it may add distance and travel time.
- Another option is to take smaller bypass roads available through local navigation services to circumvent traffic jams.

6. Local Regulations Affecting Hazardous Materials Transport:

- Transporters of hazardous materials should comply with local guidelines which may include specific permits, signage requirements, and restrictions on travel during peak movement times or in densely populated areas.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- While specific incident data may be limited, common issues have included vehicle breakdowns on poor road sections and incidents at busy intersections due to misjudged maneuvers.

8. Environmental Considerations and Sensitive Areas:

- The route may pass through agricultural zones where any spillage of hazardous material can significantly impact soil and crop quality.
- Proximity to water bodies must be considered since contamination can have widespread implications.

9. Analysis of Communication Coverage:

- Most of the route has good cellular coverage, though there might be occasional dead zones, particularly in more rural stretches.
- Ensuring a reliable communication device with GPS capability is advised.

10. Estimated Emergency Response Times for Different Route Segments:

- In urban areas, response times could range from 15-30 minutes due to proximity to emergency services.
- In rural segments, response times might increase to 45 minutes to an hour due to distance and accessibility issues.

11. Overall Summary of Risk Assessment:

- **High Risk:** During adverse weather conditions and peak traffic hours, especially at major intersections.
- **Moderate Risk:** On rural roads during night travel due to reduced visibility and infrastructure limits.
- **Low Risk:** During off-peak traffic hours and dry conditions when roadways are clear of congestion.

Conclusion: Drivers need to be trained to handle dynamic weather and traffic scenarios, pay attention to road quality, and follow all regulations. Staying informed about real-time road conditions via navigation apps and maintaining regular communication with dispatch centers is essential to ensure safety and efficiency on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
3	Turn	High	11.41981, 77.88078	15 KM/Hr
4	Turn	Medium	11.41885, 77.88455	30 KM/Hr
5	Turn	Medium	11.41829, 77.88568	30 KM/Hr
6	Turn	High	11.41859, 77.88692	15 KM/Hr
7	Turn	Medium	11.41761, 77.88739	30 KM/Hr
8	Turn	Medium	11.41716, 77.88935	30 KM/Hr
9	Turn	Medium	11.41603, 77.89479	30 KM/Hr
10	Turn	Medium	11.41439, 77.90025	30 KM/Hr
11	Turn	High	11.41377, 77.90073	15 KM/Hr
12	Turn	Medium	11.41375, 77.90128	30 KM/Hr
13	Turn	Medium	11.41389, 77.90161	30 KM/Hr
14	Turn	Medium	11.41325, 77.90313	30 KM/Hr
15	Turn	Medium	11.41276, 77.90349	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Turn	High	11.41230, 77.90485	15 KM/Hr
17	Turn	Medium	11.41262, 77.90506	30 KM/Hr
18	Turn	Medium	11.41273, 77.90524	30 KM/Hr
19	Turn	Medium	11.41329, 77.90978	30 KM/Hr
20	Turn	High	11.41424, 77.91433	15 KM/Hr
21	Turn	Medium	11.41409, 77.91458	30 KM/Hr
22	Turn	High	11.41208, 77.91551	15 KM/Hr
23	Turn	Medium	11.41218, 77.91651	30 KM/Hr
24	Turn	Medium	11.41186, 77.91768	30 KM/Hr
25	Turn	Medium	11.41342, 77.92087	30 KM/Hr
26	Turn	Medium	11.41341, 77.92113	30 KM/Hr
27	Turn	Medium	11.41319, 77.92144	30 KM/Hr
28	Turn	Medium	11.40833, 77.92362	30 KM/Hr
29	Turn	Medium	11.40669, 77.92312	30 KM/Hr
30	Turn	Medium	11.40566, 77.92355	30 KM/Hr
31	Turn	High	11.40542, 77.92346	15 KM/Hr
32	Blind Spot	Blind Spot	11.36920, 77.94568	10 KM/Hr
33	Turn	Medium	11.41332, 78.08072	30 KM/Hr
34	Turn	High	11.41463, 78.08067	15 KM/Hr
35	Turn	High	11.42964, 78.11587	15 KM/Hr
36	Turn	Medium	11.42454, 78.15291	30 KM/Hr
37	Blind Spot	Blind Spot	11.43542, 78.20754	10 KM/Hr
38	Turn	Medium	11.43040, 78.20860	30 KM/Hr
39	Turn	High	11.42171, 78.21703	15 KM/Hr
40	Turn	Medium	11.39648, 78.25107	30 KM/Hr
41	Turn	Medium	11.39541, 78.25156	30 KM/Hr
42	Turn	Medium	11.39514, 78.25153	30 KM/Hr
43	Turn	Medium	11.38226, 78.25407	30 KM/Hr
44	Turn	Medium	11.36295, 78.25903	30 KM/Hr
45	Blind Spot	Blind Spot	11.35993, 78.26200	10 KM/Hr
46	Turn	High	11.35982, 78.26194	15 KM/Hr
47	Turn	Medium	11.35851, 78.25964	30 KM/Hr
48	Blind Spot	Blind Spot	11.33761, 78.25054	10 KM/Hr
49	Turn	Medium	11.33765, 78.25083	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
50	Turn	Medium	11.33881, 78.25188	30 KM/Hr
51	Turn	Medium	11.33916, 78.25254	30 KM/Hr
52	Turn	Medium	11.33904, 78.25439	30 KM/Hr
53	Turn	Medium	11.33838, 78.25536	30 KM/Hr
54	Turn	High	11.33594, 78.25643	15 KM/Hr
55	Turn	Medium	11.33586, 78.25651	30 KM/Hr
56	Turn	High	11.33556, 78.25816	15 KM/Hr
57	Blind Spot	Blind Spot	11.33448, 78.25806	10 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
1	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
2	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
3	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41981, 77.88078



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41885, 77.88455



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41829, 77.88568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41859, 77.88692



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41761, 77.88739



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41716, 77.88935



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41603, 77.89479



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41439, 77.90025



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41377, 77.90073



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41375, 77.90128



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41389, 77.90161



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41325, 77.90313



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41276, 77.90349



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41230, 77.90485



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41262, 77.90506



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41273, 77.90524



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41329, 77.90978



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41424, 77.91433



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41409, 77.91458



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41208, 77.91551



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41218, 77.91651



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41186, 77.91768



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41342, 77.92087



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41341, 77.92113



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41319, 77.92144



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.40833, 77.92362



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.40669, 77.92312



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.40566, 77.92355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40542, 77.92346



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.36920, 77.94568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41332, 78.08072



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41463, 78.08067



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42454, 78.15291



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.43542, 78.20754



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43040, 78.20860



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42171, 78.21703



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39648, 78.25107



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39541, 78.25156



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39514, 78.25153



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38226, 78.25407



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36295, 78.25903



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.35993, 78.26200



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.35982, 78.26194



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.35851, 78.25964



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.33761, 78.25054



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33765, 78.25083



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.33448, 78.25806